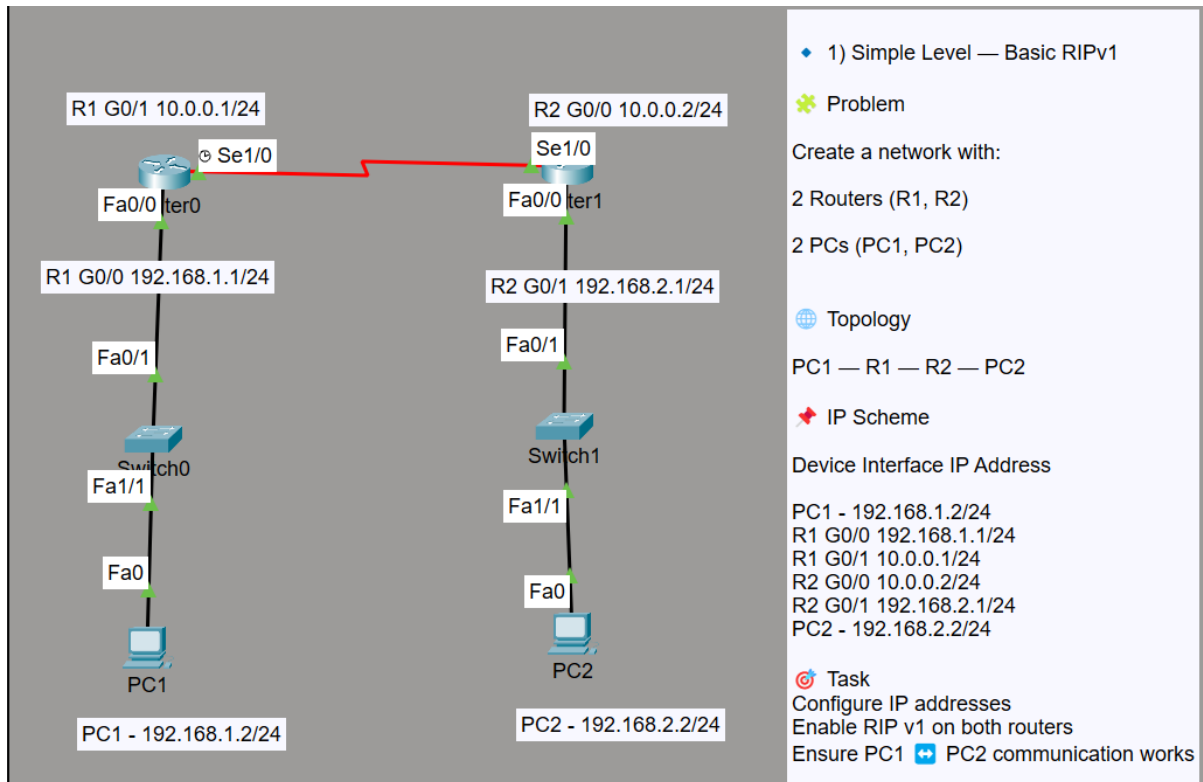


IPv4 Routing

Static

RIP v1



R1

```
Router>en
Router#config t

Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#exit

Router(config)#int se1/0
Router(config-if)#no shutdown
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#exit
```

R2

```
Router>en
Router#config t

Router(config)#int se1/0
Router(config-if)#no shut
Router(config-if)#ip add 10.0.0.2 255.255.255.0
Router(config-if)#exit

Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#ip add 192.168.2.1 255.255.255.0
Router(config-if)#exit
```

```
Router(config)#router rip
```

```
Router(config-router)#network
```

```
Router(config)#router rip
Router(config-router)#version 1
Router(config-router)#network 192.168.1.0
Router(config-router)#network 10.0.0.0
```

```
Router(config)#router rip
Router(config-router)#network 10.0.0.0
Router(config-router)#network 192.168.2.0
```

TO VERIFY

```
Router#show ip route
```

```
Router#show ip route
```

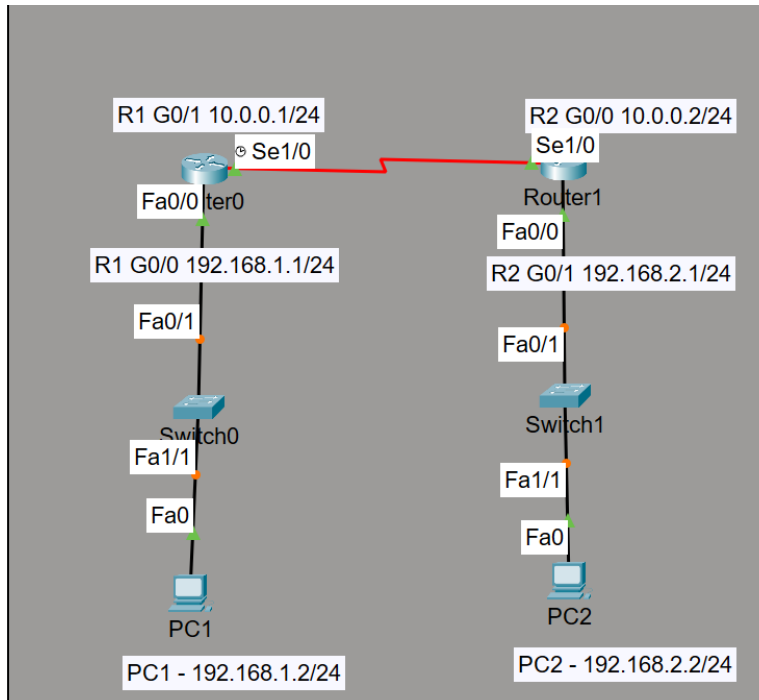
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/24 is directly connected, Serial1/0
L    10.0.0.1/32 is directly connected, Serial1/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, FastEthernet0/0
L    192.168.1.1/32 is directly connected, FastEthernet0/0
R    192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:14, Serial1/0
```

```
Router#
```

RIP v2



1) Simple Level — Basic RIPv2

Problem

Create a network with:

2 Routers (R1, R2)

2 PCs (PC1, PC2)

Topology

PC1 — R1 — R2 — PC2

IP Scheme

Device Interface IP Address

PC1 - 192.168.1.2/25
 R1 G0/0 192.168.1.1/25
 R1 G0/1 10.0.0.1/24
 R2 G0/0 10.0.0.2/24
 R2 G0/1 192.168.2.1/25
 PC2 - 192.168.2.2/25

R1

```
Router>en
Router#config t

Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#exit

Router(config)#int se1/0
Router(config-if)#no shutdown
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#exit
```

```
Router(config-if)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.1.0
Router(config-router)#exit
```

R2

```
Router>en
Router#config t

Router(config)#int se1/0
Router(config-if)#no shut
Router(config-if)#ip add 10.0.0.2 255.255.255.0
Router(config-if)#exit

Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#ip add 192.168.2.1 255.255.255.0
Router(config-if)#exit
```

```
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.2.0
Router(config-router)#network 10.0.0.0
```

TO VERIFY

Router#show ip route

Router(config)#do sh ip ro

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

C 10.0.0.0/24 is directly connected, Serial1/0

L 10.0.0.1/32 is directly connected, Serial1/0

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/25 is directly connected, FastEthernet0/0

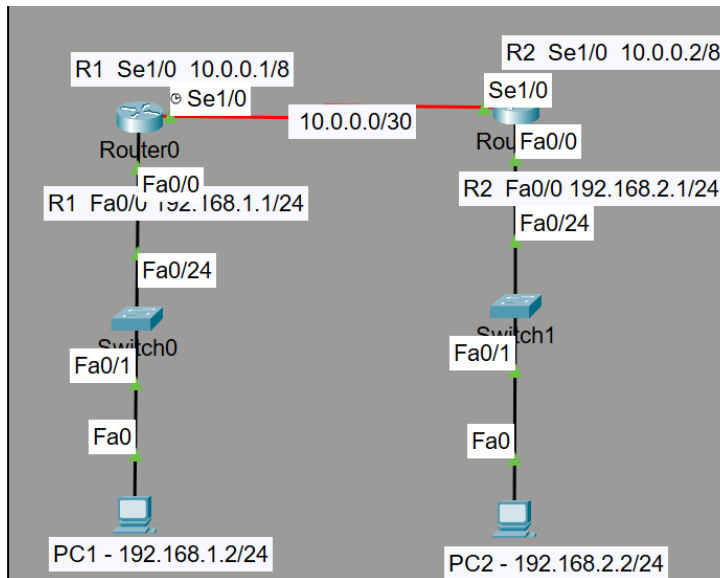
L 192.168.1.1/32 is directly connected, FastEthernet0/0

192.168.2.0/25 is subnetted, 1 subnets

R 192.168.2.0/25 [120/1] via 10.0.0.2, 00:00:05, Serial1/0

Router(config)#

EIGRP (no subnetting)



1) EIGRP — Simple Question

Problem

Create a network with:

2 Routers (R1, R2)

2 LANs

Topology

PC1 — R1 — R2 — PC2

Task

Configure IP addressing

Enable EIGRP (AS 100) on both routers

Advertise all networks

R1

Router>enable

Router#conf t

```
Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Router(config-if)#ex
```

```
Router(config)#int se1/0
Router(config-if)#no shut
Router(config-if)#ip add 10.0.0.1 255.0.0.0
Router(config-if)#exit
```

```
Router(config)#router eigrp 100
Router(config-router)#network 10.0.0.0 0.255.255.255
Router(config-router)#network 192.168.1.0 0.0.0.255
```

R2

Router>en

Router#config t

```
Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#ip add 192.168.2.1 255.255.255.0
Router(config-if)#exit
```

```
Router(config)#int se1/0
Router(config-if)#no shut
Router(config-if)#ip add 10.0.0.2 255.0.0.0
Router(config-if)#exit
```

```
Router(config)#router eigrp 100
Router(config-router)#network 10.0.0.0 0.255.255.255
Router(config-router)#network 192.168.2.0 0.0.0.255
```

TO VERIFY

Router#sh ip route

Router#sh ip route eigrp

Router#sh ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

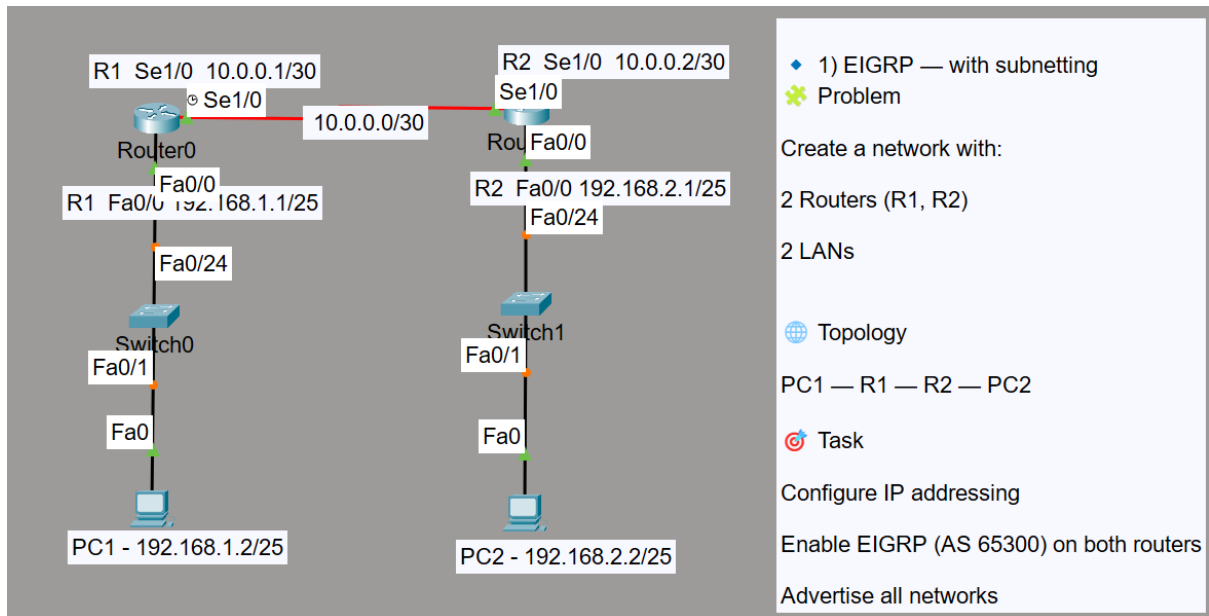
```
C 10.0.0.0/8 is directly connected, Serial1/0
D 192.168.1.0/24 [90/2172416] via 10.0.0.1, 00:01:44, Serial1/0
C 192.168.2.0/24 is directly connected, FastEthernet0/0
```

Router#sh ip route eigrp

```
D 192.168.1.0/24 [90/2172416] via 10.0.0.1, 00:02:10, Serial1/0
```

Router#

EIGRP (with subnetting)



R1

```
Router>enable
```

```
Router#conf t
```

```
Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#ip add 192.168.1.1 255.255.255.128
Router(config-if)#ex
```

```
Router(config)#int se1/0
Router(config-if)#no shut
Router(config-if)#ip add 10.0.0.1 255.255.255.252
Router(config-if)#exit
```

```
Router(config)#router eigrp 65300
Router(config-router)#network 10.0.0.0 0.0.0.3
Router(config-router)#network 192.168.1.0 0.0.0.127
Router(config-router)#exit
```

R2

```
Router>en
```

```
Router#config t
```

```
Router(config)#int fa0/0
Router(config-if)#no shut
Router(config-if)#ip add 192.168.2.1 255.255.255.128
Router(config-if)#exit
```

```
Router(config)#int se1/0
Router(config-if)#no shut
Router(config-if)#ip add 10.0.0.2 255.255.255.252
Router(config-if)#exit
```

```
Router(config)#router eigrp 65300
Router(config-router)#network 10.0.0.0 0.0.0.3
Router(config-router)#network 192.168.2.0 0.0.0.127
Router(config-router)#exit
```

TO VERIFY

Router#sh ip route

Router#sh ip route eigrp

Router#sh ip ro

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

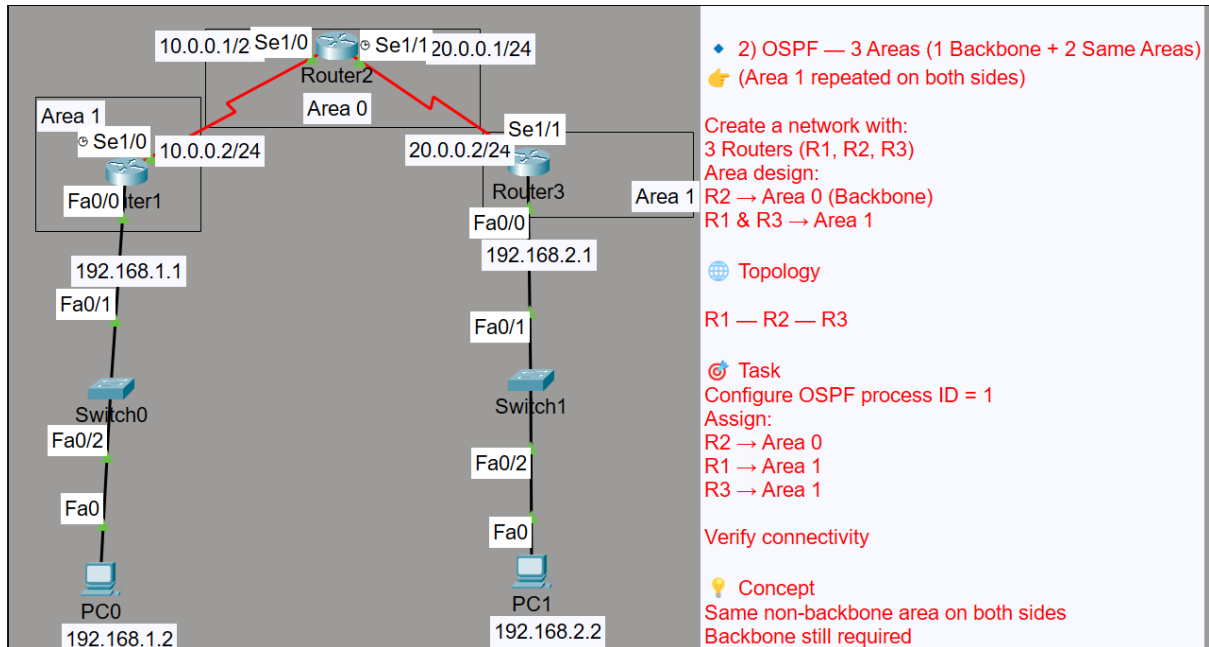
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
D 10.0.0.0/8 is a summary, 00:00:24, Null0
C 10.0.0.0/30 is directly connected, Serial1/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
D 192.168.1.0/24 is a summary, 00:00:24, Null0
C 192.168.1.0/25 is directly connected, FastEthernet0/0
D 192.168.2.0/24 [90/2172416] via 10.0.0.2, 00:00:22, Serial1/0

Router#sh ip ro eigrp

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
D 10.0.0.0/8 is a summary, 00:00:50, Null0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
D 192.168.1.0/24 is a summary, 00:00:50, Null0
D 192.168.2.0/24 [90/2172416] via 10.0.0.2, 00:00:48, Serial1/0

OSPF

Q1



```
r1(config)#router ospf 35
r1(config-router)#network 192.168.1.0 0.0.0.255 area 1
r1(config-router)#network 10.0.0.0 0.0.0.255 area 0
r1(config-router)#exit
```

```
r2(config)#router ospf 35
r2(config-router)#network 20.0.0.0 0.0.0.255 area 0
r2(config-router)#network 10.0.0.0 0.0.0.255 area 0
r2(config-router)#exit
```

```
r3(config)#router ospf 35
r3(config-router)#network 192.168.2.0 0.0.0.255 area 1
r3(config-router)#network 20.0.0.0 0.0.0.255 area 0
r3(config-router)#exit
```

TO VERIFY

```
r1#sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
```

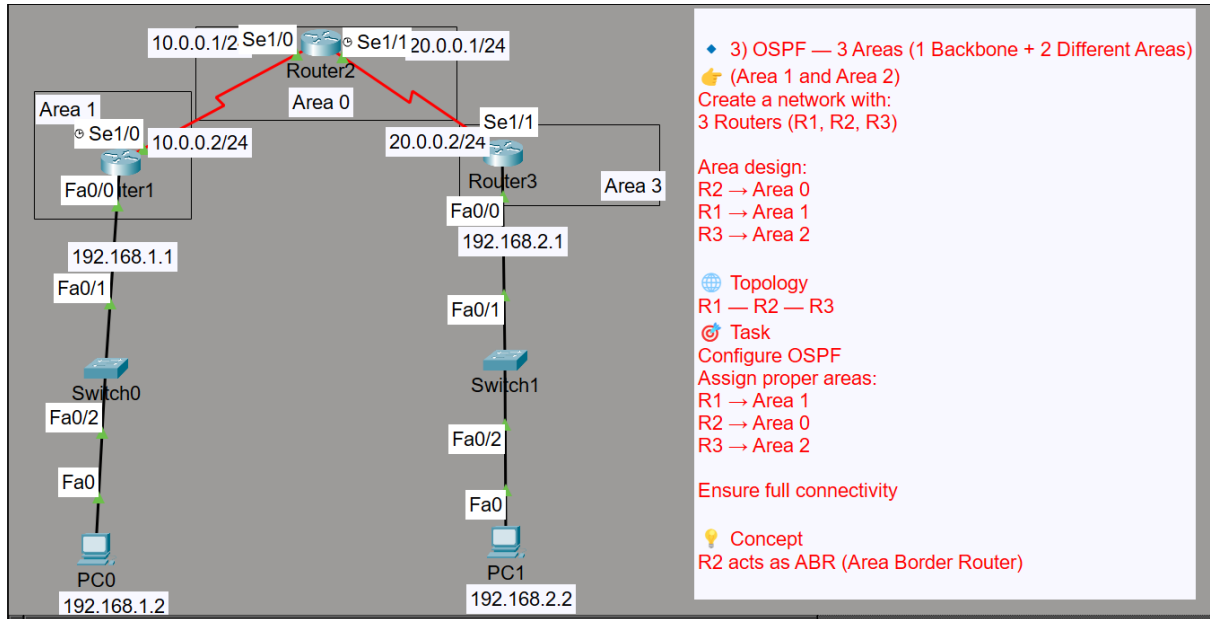
Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/24 is directly connected, Serial1/0
L 10.0.0.2/32 is directly connected, Serial1/0
20.0.0.0/24 is subnetted, 1 subnets
O 20.0.0.0/24 [110/128] via 10.0.0.1, 00:02:52, Serial1/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected, FastEthernet0/0
L 192.168.1.1/32 is directly connected, FastEthernet0/0
O IA 192.168.2.0/24 [110/129] via 10.0.0.1, 00:01:20, Serial1/0
```

r1#

Realtime Sin

Q2)



```
r1(config)#ROUTER OSPF 500
r1(config-router)#NETWORK 10.0.0.0 0.0.0.255 AREA 0
r1(config-router)#
r1(config-router)#NETWORK 192.168.1.0 0.0.0.255 AREA 1
r1(config-router)#EXIT
```

```
r2(config)#NO ROUTER OSPF 500
r2(config)#ROUTER OSPF 500
r2(config-router)#NETWORK 10.0.0.0 0.0.0.255 AREA 0
r2(config-router)#NETWORK 20.0.0.0 0.0.0.255 AREA 0
r2(config-router)#EX
```

```
r3(config)#ROUTER OSPF 500
r3(config-router)#NETWORK 192.168.2.0 0.0.0.255 AREA 3
r3(config-router)#NETWORK 20.0.0.0 0.0.0.255 AREA 0
r3(config-router)#EX
```

TO VERIFY

```
r1#SH IP RO
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
 I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
 * - candidate default, U - per-user static route, o - ODR
 P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/24 is directly connected, Serial1/0
L    10.0.0.2/32 is directly connected, Serial1/0
20.0.0.0/24 is subnetted, 1 subnets
O    20.0.0.0/24 [110/128] via 10.0.0.1, 00:03:03, Serial1/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, FastEthernet0/0
L    192.168.1.1/32 is directly connected, FastEthernet0/0
O IA 192.168.2.0/24 [110/129] via 10.0.0.1, 00:00:28, Serial1/0
```

me: 0

Realtime Simulation

Scenario 0	Fire	Last Status	Source	Destina
New			Router2	PC0
Delete			PC1	Route

Q3)

```
r1(config)#router ospf 10
```

```
r1(config-router)#network 192.168.1.0 0.0.0.255 area 2
r1(config-router)#network 10.0.0.0 0.0.0.255 area 2
r1(config-router)#exit
```

```
r2(config)#router ospf 10
```

```
r2(config-router)#network 10.0.0.0 0.0.0.255 area 2
r2(config-router)#network 20.0.0.0 0.0.0.255 area 2
r2(config-router)#exit
```

```
r3(config)#router ospf 10
```

```
r3(config-router)#network 20.0.0.0 0.0.0.255 area 2
r3(config-router)#network 192.168.2.0 0.0.0.255 area 2
r3(config-router)#exit
```

TO VERIFY

```
r2#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/24 is directly connected, Serial1/0
L    10.0.0.1/32 is directly connected, Serial1/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/24 is directly connected, Serial1/1
L    20.0.0.1/32 is directly connected, Serial1/1
O    192.168.1.0/24 [110/65] via 10.0.0.2, 00:06:18, Serial1/0
O    192.168.2.0/24 [110/65] via 20.0.0.2, 00:03:53, Serial1/1
```